

EXPERIMENT 21. Syntax-Driven Semantic Analysis

Aim: To perform syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

Program:

```
import spacy

nlp = spacy.load("en_core_web_sm")

sentence = input("Enter a sentence: ")

doc = nlp(sentence)

print("\nNoun Phrases and Meanings:")

for chunk in doc.noun_chunks:

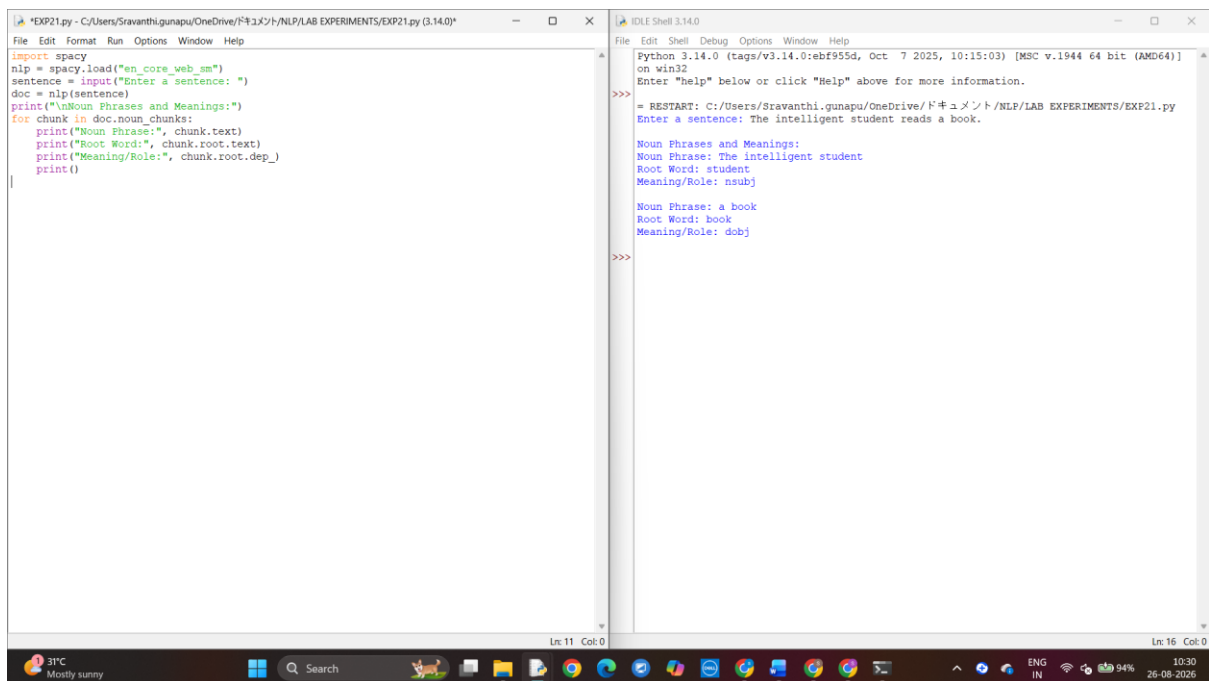
    print("Noun Phrase:", chunk.text)

    print("Root Word:", chunk.root.text)

    print("Meaning/Role:", chunk.root.dep_)

    print()
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'EXP21.py', contains the following code:

```
import spacy
nlp = spacy.load("en_core_web_sm")
sentence = input("Enter a sentence: ")
doc = nlp(sentence)
print("\nNoun Phrases and Meanings:")
for chunk in doc.noun_chunks:
    print("Noun Phrase:", chunk.text)
    print("Root Word:", chunk.root.text)
    print("Meaning/Role:", chunk.root.dep_)
    print()
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution output for the input sentence 'The intelligent student reads a book':

```
>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP21.py
Enter a sentence: The intelligent student reads a book.

Noun Phrases and Meanings:
Noun Phrase: The intelligent student
Root Word: student
Meaning/Role: nsubj

Noun Phrase: a book
Root Word: book
Meaning/Role: dobj

>>>
```

The Windows taskbar at the bottom shows the system clock as 10:30 on 26-08-2025, with a weather widget indicating 31°C and 'Mostly sunny'.